Data Analysis and Storytelling

Regression analysis - hierarchical models I

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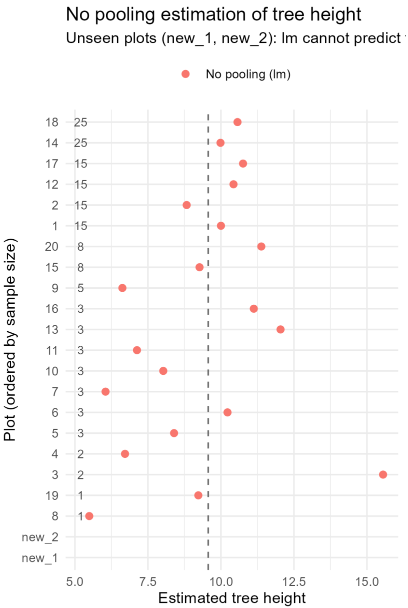
Hierarchical structures

Hierarchical structures are widespread in regression analysis:

- Trees within plots
- Plots within regions
- Birds observed by different observers

A naive approach to take into account these structure is to have a discrete covariable. This is also called ”no pooling”.

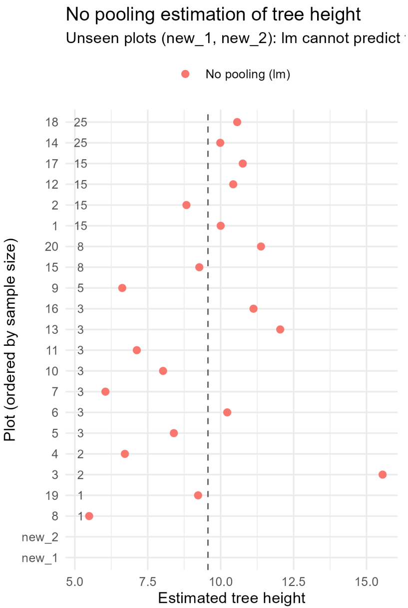


Limits of no pooling

No pooling models have some limits:

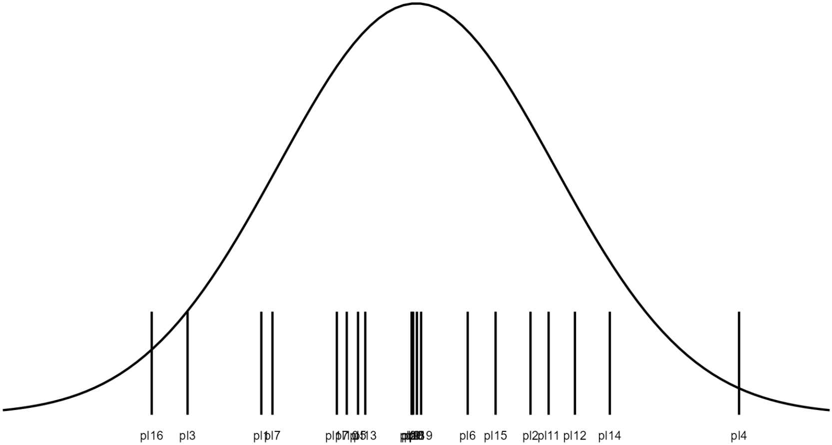
- Inefficient when the number of level becomes large (> 10)
- Does not use information across levels
- Does not allow for prediction to new level

Hierarchical models solve these issues



Intuition of hierarchical models

The values of the different levels (i.e. plots) are sampled from a distribution



A simple hierarchical model

In equation the simplest hierarchical model is:

$$y_{ij} \sim N(\mu_{ij}, \sigma_{res})$$

$$\mu_{ij} = \beta_0 + \gamma_j$$

$$\gamma_j \sim N(0, \sigma_{hier})$$

Where *j* is an indicator for the level ID (i.e. plot ID). This model is also often called a random-intercept model.

A simple hierarchical model - example

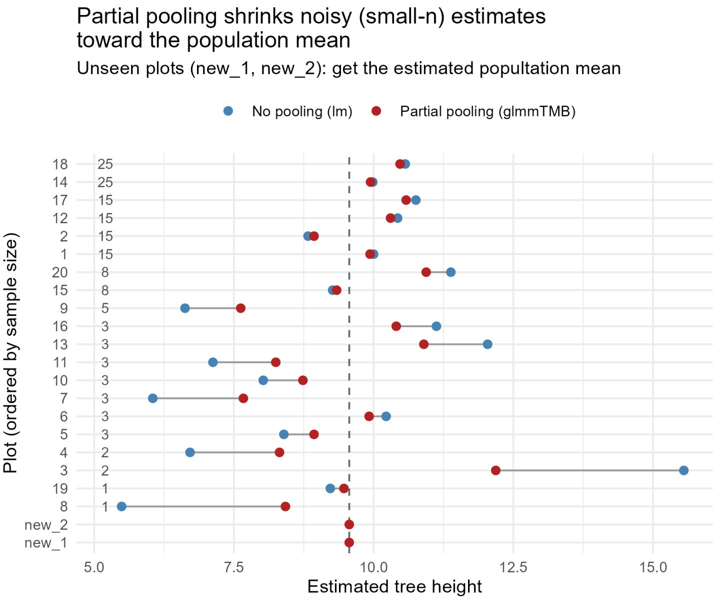
Say that we measured tree heights in different plots. The model saw previously can be written:

$$height_{tree\_id,plot\_id} \sim N(\mu_{tree\_id,plot\_id}, \sigma_{res})$$

$$\mu_{tree\_id,plot\_id} = \beta_0 + \gamma_{plot\_id}$$

$$\gamma_{plot\_id} \sim N(0, \sigma_{hier})$$

A simple hierarchical model - example



Hierarchical structure as plug-in

Hierarchical structures can be plugged in to any regression models, such as:

$$\mu_{ij} = \beta_0 + \beta_1 * X_1 + \beta_2 * X_2 + \beta_3 * X_2 * X_3 + \gamma_j$$

And this can be applied to any statistical distributions:
$y_{ij} \sim N(\mu_{ij}, \sigma)$ or $y_{ij} \sim P(\log(\mu_{ij}))$ or $y_{ij} \sim Bern(logit(\mu_{ij}))$
Think back of regression models as piling building blocks.

More hierarchies

Different hierarchical terms can be added to a regression model. For examples:

- The plots can be measured from different regions, we can expand the model we saw previously: $\mu_{ijk} = \beta_0 + \gamma_j + \delta_k$, to consider regional-level variation in tree height.
- The observer measuring tree height can vary, we can account for potential observer-level variation in measured tree heights: $\mu_{ijk} = \beta_0 + \gamma_j + \delta_k$

What makes a good hierarchical term

To choose whether a discrete covariable should be introduced as hierarchical term, various criteria can be used:

- How many levels? When the number of levels becomes large (< 10) hierarchical terms are more appropriate. In the other direction, literature recommend at least 5 levels for a hierarchical term, below this a standard covariate should be preferred.
- Are there variable sample size per group? Hierarchical models allow information to be shared across level and partial pooling prevents levels with low sample size to have too extreme values.
- Are predictions from new unobserved levels needed? Hierarchical models allow predictions from unobserved levels.

Hierarchical models and pseudoreplication

Remember that one requirement of regression model is that observations are independent from each other.

Non-independence leading to overly optimistic confidence intervals (pseudoreplication).

Hierarchical models are a straightforward way to account for this.

Dependence factors can be added to the model, such as plots, regions, observer ...